21/6/'26

Activity - 1

Q1. Medical Image Enhancement (X-ray)

* Recommended Transformation:
  Constrast stretching (Linear Gray-level Transformation)

* Justification
  → In X-ray image, pixel intensities are not uniformly distributed.
  → Bone regions are bright, while soft tissues fall in narrow low intensity range
  → Constrast stretching expands the gray-level range using:

$$s = \frac{(r - r_{min})}{(r_{max} - r_{min})} \times (L - 1).$$

  → This transformation improves dynamic range utilization of the image

* Effect on image
  → Enhance visibility of soft tissues
  → Improve contrast b/w structures
  → Preserves important features like bones
  → Produces a balanced and diagnositical useful image

Q2. Night - Time Surveillance

* Recommended Transformation

Powerlaw transformation (Gamma correction $r<1$)

* Justification

* Image pixels are concentrated in low gray-level range (0-40)
* Gamma transformation is

$$s = c \cdot r^{r}$$

→ for $r<1$ the transformation
- Expand dark pixel value
- compress brighter pixel value

→ Negative transformation is not suitable because
- It only reverses intensity values
- Does not enhance visibility of dark region

* Effect on image

→ Dark regions become brighter
→ Hidden objects become visible
→ Improve scene interpretability
→ Slight increase in noise visibility

Q3. F

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Q3. Fingerprint Recognition

* Recommended Transformation

  Histogram Equalization

* Justification

  → Fingerprint images requires enhancement of fine ridge and valley pattern

  → Histogram equalization:

  - Redistribution pixel intensities

  $$s = (L-1)\sum_{i=0}^{r} P_r(i)$$

  → Improve global contrast

* Log Transformation

  → Log transformation

  $$s = c\log(1+r)$$

  → Enhance only low-intensity region

  → Less effective compared to histogram equalization.